Fei Hu

Curriculum Vitae

140 Maryland Hall
Johns Hopkins University
Baltimore, MD 21218
fhu13@jh.edu

EDUCATION & TRAINING

Johns Hopkins University — Baltimore, MD
Major: Materials Science and Engineering (Postdoctoral fellow) — Mar. 2025 – Present
- Advisor: Prof. Yuting Luo

University of Rochester — Rochester, NY
Major: Chemical Engineering (**Ph.D.**) — Aug. 2020 – Jan. 2025
- Advisor: Prof. Wyatt E. Tenhaeff
- Thesis: Development of thin-film materials to enhance the energy density, lifespan and safety of Lithium batteries

University of Rochester — Rochester, NY
Major: Chemical Engineering (**M.S.**) — Sep. 2018 – May 2020
- Advisor: Prof. Wyatt E. Tenhaeff

South China University of Technology — Guangzhou, China
Major: Energy Chemical Engineering (**B.Eng.**) — Sep. 2014 – June 2018
Double Major: Business Administration
- Advisor: Prof. Xueqing Qiu
- Thesis: Preparation of SnO_2/C composites as Li ion battery anodes and investigation of their lithiation performance

FELLOWSHIPS & AWARDS

Johns Hopkins University — Baltimore, MD
Ralph O'Connor Sustainable Energy Institute (ROSEI) Postdoctoral Scholars — July 2025 – July 2026
- $60,000 in stipend support

University of Rochester — Rochester, NY
Ph.D. Achievement Award — May 17th, 2024
- $300

University of Rochester — Rochester, NY
Elon Huntington Hooker Dissertation Fellowship — July 2023 – June 2024
- $24,700 in stipend support

University of Rochester — Rochester, NY
Kwang-Yu and Lee-Chien Wang Graduate Fellowship — Feb. 2023 – Dec. 2023
- $12,000 in research funds
- $2,000 in academic conferences travel grant
- $16,000 in academic stipend

241st ECS Meeting — Vancouver, Canada

Travel Grant May 29th – June 2nd, 2022

- $290

University of Rochester Rochester, NY

Earl W. Costich Graduate Fellowship Sep. 2021 – May 2022

- $1800

2020 Virtual MRS Fall Meeting & Exhibit Boston, MA

Best Presentation Award (Energy Symposium) Nov. 27th – Dec. 4th, 2020

- $250

TEACHING & MENTORING EXPERIENCE

Johns Hopkins University Baltimore, MD

Student Mentoring

Andrea Hernandez (summer research undergraduate from California Polytechnic State University)

May 2025 – July. 2025

- Awarded outstanding poster design at PROPEL Research Symposium 2025

University of Rochester Rochester, NY

Student Mentoring

Suganesh Raghunathan (Materials Science, master student) Apr. 2024 – Jan. 2025

Adel Fadhul (Chemical Engineering, Class of 2025) Jun. 2023 – Aug. 2023

- Awarded 2nd place student poster (Fuels, Petrochemicals, and Energy II) at AIChE (2023)

Rebecca Choi (Chemical Engineering, Class of 2024) Jun. 2022 – Oct. 2022

- Awarded 3rd place student poster at NY BEST (2022)
- Eisenberg research fellow
- Full-time paid summer internship at Oak Ridge National Laboratory

Teaching Assistant---*Advanced Transport Phenomena* Sep. 2021 – Dec. 2021

- Worked with professor to revise group project and graded homework
- Held weekly two-hour office hours to discuss projects and assignments

Teaching Assistant---*Solar Cells* Sep. 2019 – Dec. 2019

- Worked with professor to revise group project and graded midterm exam
- Lead teaching fellow team consisting of 3 graduate student colleagues
- Held weekly two-hour office hours to discuss projects and assignments
- Average grade of the class hit a new record high

PUBLICATIONS

9. **F. Hu**, Z. Li and W.E. Tenhaeff, "Monitoring Temperature Evolution of Batteries". (In preparation)
8. **F. Hu**, Y. Zhao, X Li, IW Yeu, A Urban and W.E. Tenhaeff, "Combining Ultrathin Lithiophobic and Lithiophilic Interlayers to Enhance the Reversibility of Anode-Free Lithium Metal Batteries". ***Energy Storage Materials***, 2026, 104883.
7. **F. Hu**‡, Z. Ye‡, A.D. Pickel and W.E. Tenhaeff, "Operando Characterization of Lithium Batteries Internal Temperature via Upconverting Nanoparticle Thermometry", ***Nanoscale***, 2025, 17, 14597-14606. (‡Contributed equally)
6. Y. Zhao‡, **F. Hu**‡ and W.E. Tenhaeff, "Control of Fluoropolymer Crystallinity for Flexible, Transparent Optical Thin Films with Low Refractive indexes", ***Macromol.***, 2025, 58, 3, 1265-1278. (‡Contributed equally)

5. **F. Hu**, Z. Li, W.E. West and W.E. Tenhaeff, "High-Capacity, Nanocrystalline Li_2RuO_3-$LiCoO_2$ Cathodes for Flexible Solid-State Thin Film Battery", ***J. Power Sources***, 2022, 551, 232206.
4. Z. Li, **F. Hu**, N. Huo and W.E. Tenhaeff, "A Flexible Thin Film Lithium Battery with Chemical Vapor Deposited Organic Complex Cathode", ***J. Mater. Chem. A***, 2022, 10, 8390-8400.
3. **F. Hu**, Z. Li, S. Wang and W.E. Tenhaeff, "Mirror-Like Electrodeposition of Lithium Metal under a Low-Resistance Artificial Solid Electrolyte Interphase Layer", ***ACS Appl. Mater. & Interfaces***, 2020, 12 (35), 39674-39684.
2. M.M. Loanniti, **F. Hu** and W.E. Tenhaeff, "Energy-Dense Li Metal Anodes Enabled by Thin Film Electrolytes", ***J. Vac. Sci. Technol. A***, 2020, 38 (6), 060801.
1. R. Zou, J. Zhang, S. Hu, **F. Hu**, H. Zhang and Z. Fu. "Reversible Mechanochromic and Thermochromic Luminescence Switching via Hydrogen-Bond-Directed Assemblies in a Zinc Coordination Complex", ***CrystEngComm.***, 2017, 19 (42), 6259-6262.

INTELLECTUAL PROPERTIES

3. **F. Hu**, W.E. Tenhaeff, W.C. West, "High Capacity Cathodes for All-Solid-State Thin-Film Batteries", **U.S. Patent**, 2022, 17/964, 746. (Pending)
2. **F. Hu**, "Thermochemical Energy Storage Reaction Device and Application System Thereof", **China Patent for Utility Models**, 2020, CN 212645465U.
1. **F. Hu**, "Thermochemical Energy Storage Reaction Device, Energy Storage Method and Application System Thereof", **China Patent for Invention**, 2020, CN 111288828A.

PEER REVIEW

Journal Reviewer

- Vacuum

PRESENTATIONS & POSTERS

15. A. Hernandez, **F. Hu** and Y. Luo, "Engineering Metal Coatings for Safer Anode-free Solid-State Batteries", ***2025 Promoting Research Opportunities in Engineering Labs (PROPEL) Poster Session***, Baltimore, MD, July 2025. (Poster)
14. A. Fadhul, **F. Hu** and W.E. Tenhaeff, "MgO as a Water Scavenger in Lithium-Ion Batteries' Separators", ***2023 AIChE Annual Meeting***, Orlando, FL, Nov. 2023. (Poster)
13. **F. Hu** and W.E. Tenhaeff, "Multifunctional Separator for Li-Ion Batteries Enabling Acid Scavenging and Long Cyclability", ***244th ECS Meeting***, Gothenburg, Sweden, Oct. 2023. (Oral Presentation)
12. **F. Hu**, Z. Li, and W.E. Tenhaeff, "High-Capacity Cathodes for Flexible Thin-Film Batteries", ***Poster Session in Chemical Engineering, University of Rochester***, Rochester, NY, Dec. 2022. (Poster)
11. **F. Hu** and W.E. Tenhaeff, "Development of Thin Film Materials for Li Batteries", ***Research Seminar in Chemical Engineering, University of Rochester***, Rochester, NY, Nov. 2022. (Oral Presentation)
10. **F. Hu**, Z. Li, W.C. West and W.E. Tenhaeff, "High-Capacity Cathodes for Flexible Thin-Film Batteries", ***NY-BEST Annual Fall Energy Storage Technology & Innovation Conference***, Rochester, NY, Oct. 2022. (Poster)
9. **F. Hu** and W.E. Tenhaeff, "Thin Film Composite Separators to Enable Li Metal Anodes", ***ACS NERM 2022***, Rochester, NY, Oct. 2022. (Oral Presentation)
8. **F. Hu**, Z. Li, W.C. West and W.E. Tenhaeff, "High-Capacity Cathodes for All-Solid-State Thin-Film Batteries", ***241st ECS Meeting***, Vancouver, Canada, May 2022. (Oral Presentation)
7. **F. Hu**, Z. Li, S. Wang and W.E. Tenhaeff, ""Mirror-Like Electrodeposition of Lithium Metal under a Low-Resistance Artificial Solid Electrolyte Interphase Layer", ***239th ECS Meeting***, Chicago, IL, May 2021. (Digital Oral Presentation)
6. **F. Hu**, S. Wang, Z. Li and W.E. Tenhaeff, "Uniform Electrodeposition of Li Metal under a Low-Resistance Artificial Solid Electrolyte Interface", ***NY-BEST Energy Storage Innovation Conference***, Binghamton, NY, Dec. 2020. (Digital

Poster)

5. **F. Hu** and W.E. Tenhaeff, ""Mirror-Like" Electrodeposition of Lithium Metal under a Low-Resistance Artificial Solid Electrolyte Interphase Layer", ***2020 MRS Fall Meeting***, Boston, MA, Nov. 2020. (Digital Oral Presentation)
4. **F. Hu**, S. Wang, Z. Li and W.E. Tenhaeff, "Uniform Electrodeposition of Lithium Metal Enabled by Thin Lithium Phosphorous Oxynitride (LiPON) Films", ***2020 Virtual AIChE Annual Meeting***, San Francisco, CA, Nov. 2020. (Digital Oral Presentation)
3. **F. Hu**, M.M. Loanniti, S. Wang and W.E. Tenhaeff, "Lithium Batteries Enabled by Thin Film Composite Solid Electrolyte Separators", ***237th ECS Meeting (IMCS 2020)***, Montreal, Canada, May 2020. (Invited Talk, Cancelled)
2. **F. Hu**, Z. Li, S. Wang, Y. Zhao and W.E. Tenhaeff, "Uniform Electrodeposition of Lithium Metal Under an Artificial Solid Electrolyte Interphase Layer", ***237th ECS Meeting (IMCS 2020)***, Montreal, Canada, May 2020. (Oral Presentation, Cancelled)
1. **F. Hu**, Z. Li, S. Wang and W.E. Tenhaeff, "Study of Electrodeposited Lithium Metal Protected by a Thin Film LiPON Electrolyte", ***NY-BEST Energy Storage Innovation Conference***, Rochester, NY, Oct. 2019. (Poster)